

Improving stability and performance of s-step Krylov subspace methods with randomization

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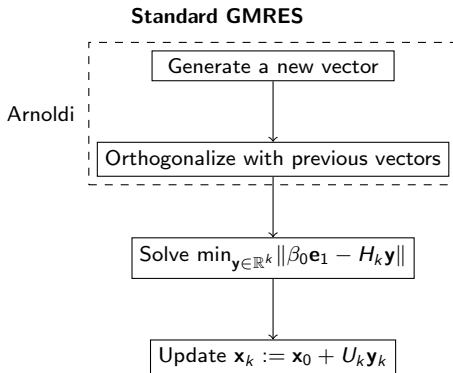
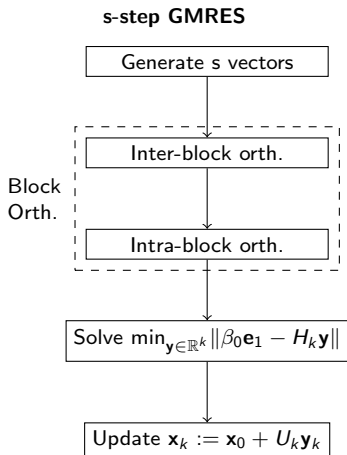
Background

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- Nowadays, with the increasing availability of high-performance computing (HPC) platforms and large-scale distributed systems, Krylov subspace methods have become an effective approach for solving large-scale linear systems. However, when deployed on modern compute clusters, the performance of these methods is often limited by the communication cost among computing nodes, which has emerged as a critical bottleneck.
- To mitigate communication overhead, some communication-avoiding techniques have been proposed (e.g. s-step Krylov method). However, the bottleneck lies in the convergence of power methods when generating multiple bases, thus it will lead to significant loss of orthogonality.
- To improve the stability and performance of s-step Krylov subspace methods, we try to introduce randomization into the orthogonalization process using subspace embeddings.

s-step GMRES vs. Standard GMRES

To solve a linear system $A\mathbf{x} = \mathbf{b}$, in the k -th iteration:



Subspace Embedding

ϵ -subspace embedding

For $\epsilon < 1$, the sketching matrix $\Theta \in \mathbb{R}^{d \times n}$ is said to be an ϵ -subspace embedding for $\mathcal{V} \subseteq \mathbb{R}^n$ if we have

$$(1 - \epsilon) \|\mathbf{v}\|^2 \leq \|\Theta \mathbf{v}\|^2 \leq (1 + \epsilon) \|\mathbf{v}\|^2, \quad \forall \mathbf{v} \in \mathcal{V}. \quad (1)$$

We aim for constructions that use little or even no information about the subspace $\mathcal{V}$, which introduces

(k, ϵ, δ) -Oblivious Subspace Embedding (OSE)

A (k, ϵ, δ) -oblivious subspace embedding (OSE) is a random matrix Θ such that, for a fixed but arbitrary k -dimensional subspace $\mathcal{V} \subset \mathbb{R}^n$, the matrix Θ is an ϵ -subspace embedding with probability at least $1 - \delta$. That is,

$$(1 - \epsilon) \|\mathbf{v}\|^2 \leq \|\Theta \mathbf{v}\|^2 \leq (1 + \epsilon) \|\mathbf{v}\|^2, \quad \forall \mathbf{v} \in \mathcal{V}.$$

Subspace Embedding

Gaussian Embedding

The Gaussian embedding is the standard Gaussian random matrix scaled by a coefficient related with sketch size d

Let

$$\Omega \in \mathbb{R}^{d \times n} \text{ with i.i.d. } \mathcal{N}(0, 1) \text{ entries,}$$

then

$$\Theta_G = \frac{1}{\sqrt{d}} \Omega.$$

Rademacher

A Rademacher matrix $R \in \mathbb{R}^{d \times n}$ is a random matrix with i.i.d. entries R_{ij} :

$$P(R_{ij} = 1) = \frac{1}{2}, \quad P(R_{ij} = -1) = \frac{1}{2}, \quad i = 1, \dots, d, \quad j = 1, \dots, n.$$

Then $\Theta_R = \frac{1}{\sqrt{d}} R$.

Sparse Sign Embedding

The sparse sign embedding is a sparse matrix with nonzero entries that are random signs has the following form

$$\Theta_S = \frac{1}{\sqrt{\xi}} [s_1, s_2, \dots, s_n] \in \mathbb{R}^{d \times n},$$

where the columns s_i of Θ_S are i.i.d. sparse vectors with exactly ξ nonzero entries whose locations are chosen uniformly at random, and whose nonzero values are i.i.d. Rademacher random variables.

Subspace Embedding

To ensure a sparse sign embedding $\Theta \in \mathbf{R}^{d \times n}$ to be oblivious for a matrix $A \in \mathbf{R}^{n \times m}$, we have the following requests for the sketch size d and the cost of matrix multiplication:

Subspace Embedding	Sketch size d	Matrix mult. cost
Gaussian	$\mathcal{O}\left(\epsilon^{-2} \left(m + \log \frac{1}{\delta}\right)\right)$	$\mathcal{O}(dnm)$
Rademacher	$\mathcal{O}\left(\epsilon^{-2} \left(m + \log \frac{1}{\delta}\right)\right)$	$\mathcal{O}(dnm)$
Sparse sign ¹	$\mathcal{O}(m)$	$\mathcal{O}(\xi nm)$

¹The theoretical choices for d and ξ are $\mathcal{O}(m \log m)$ and $\mathcal{O}(\log m)$. In practice, $d = \mathcal{O}(m)$ and $\xi = \min\{d, 8\}$ is recommended by Tropp et al., SIAM J. Sci. Comput., 41 (2019), pp. A2430–A2463.

BGS Process and RBGS Process

Orthonormalize $X \in \mathbb{R}^{n \times m} = [X_{(1)}, X_{(2)}, \dots, X_{(p)}]$ ($X_{(i)} \in \mathbb{R}^{n \times s}$) s.t. $X = QR$

- The block orthogonalization in the s-step GMRES is performed via the block Gram–Schmidt (BGS) process, a two-step procedure for producing orthonormal vectors.
- ❶ **Inter-block orthogonalization:**
orthogonalize $X_{(i)}$ against blocks $Q_{(1)}, Q_{(2)}, \dots, Q_{(i-1)}$ yielding $R(1 : (i-1)s, (i-1)s + 1 : is)$ and
 $Q'_i = X_{(i)} - Q(:, 1 : (i-1)s) R(1 : (i-1)s, (i-1)s : is)$
- ❷ **Intra-block orthogonalization:**
apply orthogonalization process to Q'_i , yielding
 $Q_{(i)} = Q(:, (i-1)s + 1 : is)$ and $R((i-1)s + 1 : is, (i-1)s + 1 : is)$

$$\Theta\text{-Orthonormalize } X = [X_{(1)}, X_{(2)}, \dots, X_{(p)}] \text{ s.t. } X = QR$$

- RBGS outputs a sketch-orthonormal factor Q (i.e., $(\Theta Q)^T \Theta Q = I$)

❶ **Inter-block Θ -orthogonalization:**

compute sketched matrix ΘX_i , orthogonalize ΘX_i with $S(:, 1 : (i-1)s)$, yielding $R(1 : (i-1)s, (i-1)s : is)$; compute Θ -orthogonalized matrix against blocks $Q_{(1)}, Q_{(2)}, \dots, Q_{(i-1)}$:

$$Q'_i = X_i - Q(:, 1 : (i-1)s)R(1 : (i-1)s, (i-1)s : is)$$

❷ **Intra-block Θ -orthogonalization:**

apply randomized orthogonalization process to Q'_i , yielding $Q_{(i)} = Q(:, (i-1)s + 1 : is)$ and $R((i-1)s + 1 : is, (i-1)s + 1 : is)$

❸ **Compute sketched matrix $S(:, (i-1)s + 1 : is) = \Theta Q(:, (i-1)s + 1 : is)$**

Randomized Orthogonalization Methods

Randomized Orthogonalization Methods

- Similar to the goal of the standard QR decomposition for a matrix $X \in \mathbb{R}^{n \times m}$, the randomized orthogonalization procedure aims to produce an **approximately orthonormal** factor Q together with an upper-triangular factor R .
- In essence, the randomized orthogonalization of a matrix $X \in \mathbb{R}^{n \times m}$ is to orthogonalize ΘX .
- It produces an upper triangular factor R and a sketch-orthonormal matrix Q (which means $(\Theta Q)^T (\Theta Q) = I$) such that $X = QR$.
- Mathematically speaking, we can define an operator

$$P_{\mathcal{X}} := Q(\Theta Q)^T \Theta.$$

to describe the whole process, here $P_{\mathcal{X}}$ is idempotent but not self-adjoint and is an oblique projection onto the subspace $\mathcal{X} \subset \mathbb{R}^n$.

Randomized Variants of standard Gram-Schmidt

We can define a sketch-orthogonal projector Ω for Randomized Variants of standard Gram-Schmidt

- For rCGS,

$$\Omega^{(j)} = I_{n \times n} - P_{\mathcal{V}_j}, P_{\mathcal{V}_j} = Q_j(\Theta Q_j)^T \Theta$$

- For rMGS,

$$\Omega^{(j)} = (I_{n \times n} - P_{q_j})(I_{n \times n} - P_{q_{j-1}}) \dots (I_{n \times n} - P_{q_1}), P_{q_j} = q_j(\Theta q_j)^T \Theta$$

```
for  $i = 1$  to  $m$  do
   $\mathbf{q}'_i = \Omega^{(i-1)} \mathbf{x}_i$ , also yielding  $R(1 : i - 1, i)$  ;
   $R(i, i) = \|\Theta \mathbf{q}'_i\|$  ;
   $\mathbf{q}_i = \mathbf{q}'_i / R(i, i)$  ;
end for
```

Randomized Gram-Schmidt (RGS)

In RGS, we compute the upper triangular factor R by solving a sketched least-square problem:

$$\begin{aligned} R(1 : i - 1, i) &= \arg \min_{\mathbf{y}} \|\Theta(Q(:, 1 : i - 1)\mathbf{y} - \mathbf{x}_i)\| \\ &= \arg \min_{\mathbf{y}} \|S(:, 1 : i - 1)\mathbf{y} - \mathbf{p}_i\| \end{aligned}$$

RGS

```
for  $i = 1$  to  $m$  do  
   $\mathbf{p}_i = \Theta \mathbf{x}_i$  ;  
   $R(1 : i - 1, i) = \arg \min_{\mathbf{y}} \|S(:, 1 : i - 1)\mathbf{y} - \mathbf{p}_i\|$  ;  
   $\mathbf{q}'_i = \mathbf{x}_i - Q(:, 1 : i - 1) * R(1 : i - 1, i)$  ;  
   $R(i, i) = \|\Theta \mathbf{q}'_i\|$  ;  
   $\mathbf{q}_i = \mathbf{q}'_i / R(i, i)$  ;  
end for
```

Randomized Cholesky QR

Randomized Cholesky QR directly computes the upper triangular factor R of the sketched matrix ΘX

rCholesky

$$\begin{aligned} P &= \Theta X ; \\ R &= qr(P) ; \\ Q &= XR^{-1} ; \end{aligned}$$

- Although these methods are all equivalent in exact arithmetic, they would have different performance when implemented finite precision arithmetic because of the different ways in accumulating rounding errors.

Randomized s-step GMRES

Randomized s-step GMRES

To solve $A\mathbf{x} = \mathbf{b}$, $A \in \mathbb{R}^{n \times m}$, $\Theta \in \mathbb{R}^{d \times n}$, $d = 2m \ll n$, step size s , iteration number p , dimension of the final Krylov subspace is $m = s * p + 1$.

Algorithm (RBGS-GMRES)

for $j = 1$ to p **do**

$i = s * (j - 1) + 1$;

$cols = (i + 1) : (i + s)$;

$V(:, cols) = \text{BasisGen}(A, Q(:, i), s)$;

Θ -orthogonalize $V(:, cols)$ with $Q(:, 1 : i)$;

Θ -orthonormalize $Q(:, cols)$;

RBGS process

$S(:, cols) = \Theta * Q(:, cols)$;

 Update Hessenberg matrix explicitly $\rightarrow H = S \setminus (\Theta A Q)$ since $AQ = QH$;

 Solve the least-square problem: $\mathbf{y} = \arg \min_{\mathbf{z}} \|H\mathbf{z} - \beta_0 \mathbf{e}_1\|$;

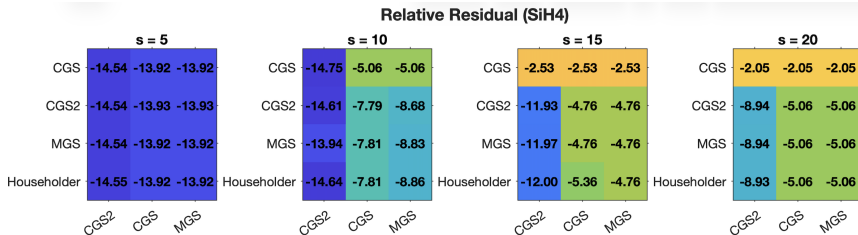
end for

 Update the solution $\rightarrow \mathbf{x} = \mathbf{x}_0 + Q(:, 1 : ps) * \mathbf{y}$;

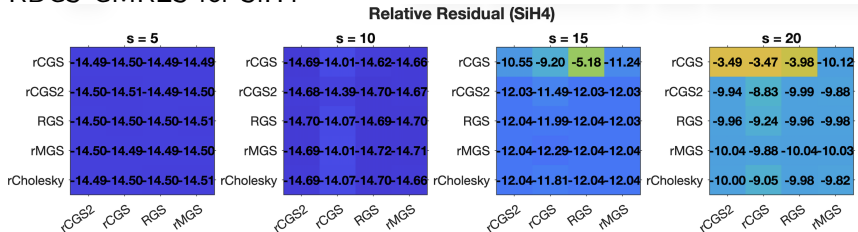
Numerical Experiments

Comparison of BGS and RBGS in GMRES

- BGS-GMRES for SiH4 (Rows: intra-block, Columns: inter-block)

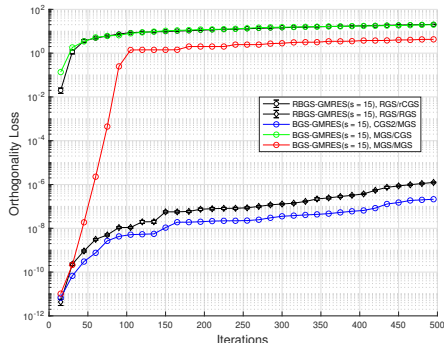
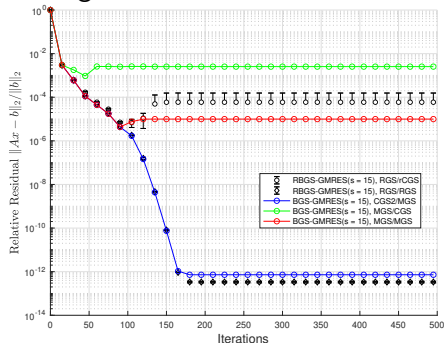


- RBGS-GMRES for SiH4



Comparison of BGS and RBGS in GMRES

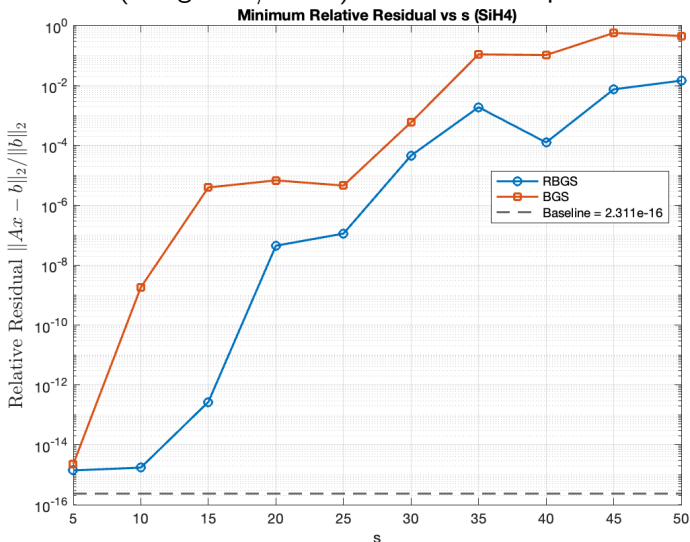
Orthogonalization Schemes for SiH4 over 20 runs when $s = 15$ ²



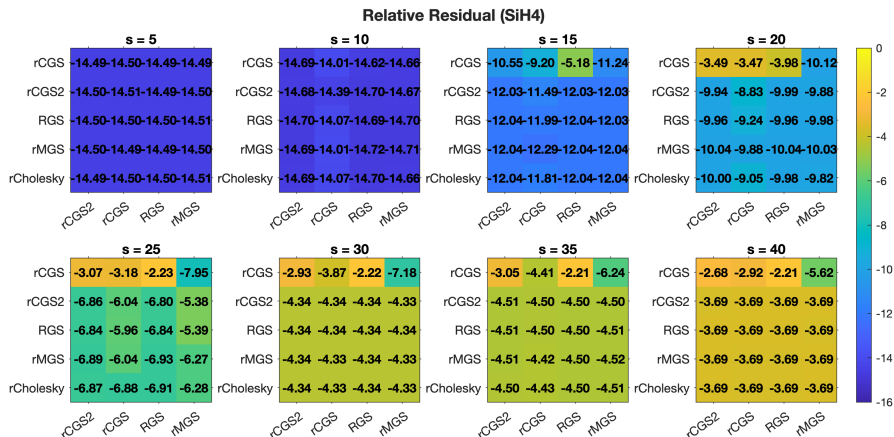
²In the legend, the first method refers to inter-block orthogonalization and the second to intra-block orthogonalization. For example, RGS/rCGS means using RGS for inter-block and rCGS for intra-block orthogonalization.

Comparison of BGS and RBGS in GMRES

Minimum relative residual achieved by RBGS-GMRES (using RGS/RGS) and BGS-GMRES (using CGS/CGS2) for different step sizes s

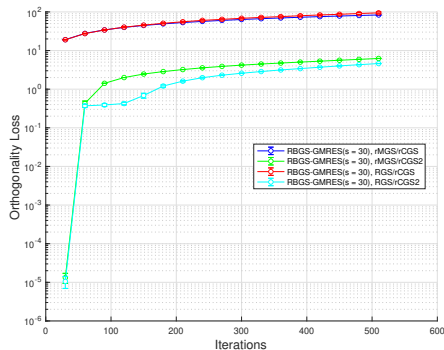
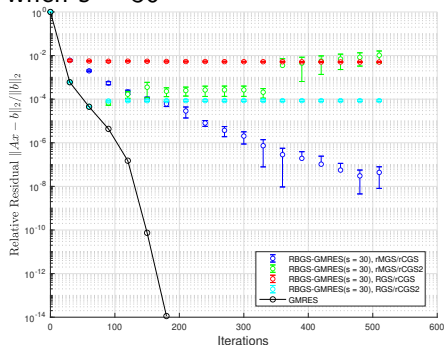


Delayed Convergence Phenomena of rMGS/rCGS in RBGS-GMRES



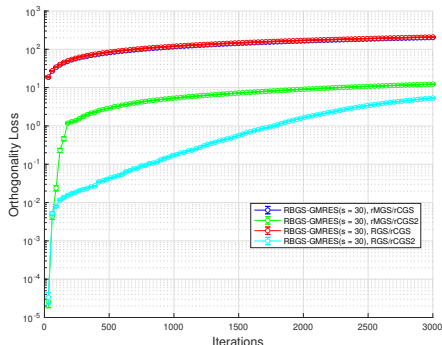
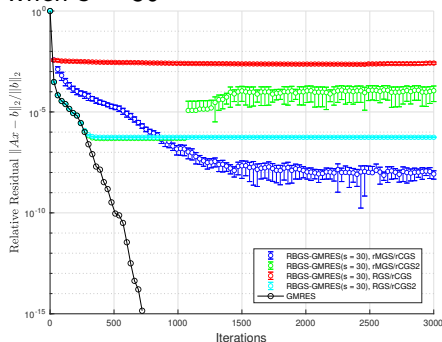
Delayed Convergence Phenomena of rMGS/rCGS in RBGS-GMRES

Different Randomized Orthogonalization Schemes for **SiH4** over 20 runs when $s = 30$



Delayed Convergence Phenomena of rMGS/rCGS in RBGS-GMRES

Different Randomized Orthogonalization Schemes for **SiO2** over 20 runs when $s = 30$



Thank you